

King Owusu (**Back Propagation**)

For this group project, my main contribution was researching and explaining backpropagation in deep learning. I focused on understanding how a neural network learns from its mistakes and how backpropagation helps it make better predictions. I learned that when a neural network makes a prediction, it compares the prediction to the correct answer to find the error. Then, it works backward through the network to figure out which weights contributed to the mistake. Backpropagation calculates gradients that show how the weights should be changed, and those changes help the network improve its future predictions.

My job was to create slides about backpropagation and explain it in a way that an 11-year-old could understand. I used a simple example of a neural network looking at a picture of a cat but incorrectly predicting that it was a dog. I explained backpropagation as looking back at a mistake to figure out what went wrong and then making changes so the same mistake is less likely to happen next time. This made me realize that explaining something complicated to a younger audience is not always easy. You have to use simple words, relatable examples, and visuals without leaving out the important information.

This project helped me understand backpropagation better and also taught me more about working in a group. While researching, I was able to see how errors, weights, and gradients all work together to help a neural network learn. I also realized that a good presentation is not just about having accurate information. You also need to explain it in a way that people can actually follow and remember. Overall, I feel that this project helped me improve both my understanding of deep learning and my ability to explain technical concepts in a simple way.

Esther Odaibo (**Entropy and Cross Entropy**)

For my part of the project, I researched entropy and cross entropy and created two slides that explain both concepts to an 11-year-old. I used a mystery-bag example for entropy. When several possible answers have similar chances, the model has high uncertainty, so entropy is high. When one answer has a much higher chance than the others, uncertainty and entropy are lower. This helped me understand that entropy measures uncertainty within the model's probability distribution. It does not determine whether the model's prediction is correct.

For cross entropy, I used a mistake-score example. The model's predicted probabilities are compared with the correct label. Giving the correct answer a high probability produces a small loss, while being confident about the wrong answer produces a much larger loss. During training, the network tries to reduce this loss. Backpropagation then helps determine how the weights contributed to the error and how they should be adjusted.

I also explained where my analogies stop being accurate. A neural network does not actually feel confused, and the cross-entropy score does not change the weights by itself. The network performs mathematical calculations using probabilities. My contribution included researching the concepts, developing the child-friendly examples, creating the two presentation slides and adding supporting sources and speaker notes. This project helped me understand how entropy describes uncertainty, while cross entropy measures prediction error and gives the network a useful learning signal.

For this project, I researched gradient descent and helped develop our title and introduction slides. I used ChatGPT as a learning partner to explore the history, question explanations, and turn what I learned into examples an 11-year-old could follow.

Because I enjoy history, I started with the person behind the method. Augustin-Louis Cauchy was a French mathematician who described a gradient-based method in 1847. He discussed applying it to improve calculations of objects' orbits in space. That connection made the topic memorable: long before teaching computers, mathematicians were finding ways to improve estimates by reducing their errors.

My biggest breakthrough came from questioning the mountain analogy. What does “downhill” actually mean? I learned that the hill's height represents loss, an error score, while moving across it represents changing the model's settings. The gradient points toward the steepest local increase in loss, so gradient descent moves in the opposite direction. Lowering loss does not mean making every weight smaller. Some settings may need to increase while others decrease, and the learning rate controls the size of the adjustment.

I also asked, “Correct compared with what?” In our cat-and-dog example, the training label supplies the answer key. A guess of “dog” is checked against the photo's “cat” label. To make weights less mysterious, we compared them to speaker-volume controls: number settings that change how strongly something comes through. Keeping the same animal example helped connect the explanation from beginning to end.

Working with ChatGPT was most useful when I pushed past a definition and asked what it actually meant. That process helped me connect Cauchy's historical problem to neural-network learning. My main takeaway is that simplifying a concept does not mean removing its meaning. It means building from something familiar until the technical idea makes sense.

Zain Ahmed (Weights, Bias and Activation Functions Reflection)

For my part of this project, I researched weights, bias, and activation functions and created two slides explaining them in a way that an 11-year-old could understand. While working on my topic, I learned that weights control how much different information matters to a neuron. Bias gives the result a small adjustment, while the activation function helps decide what information gets passed forward in the network. All three of these work together to help a neural network make predictions.

To make my topic easier to understand, I used the same cat-and-dog example that we used throughout our presentation. I explained weights as how much the computer should pay attention to different clues, bias as a small adjustment to the score, and the activation function as what happens to that score before it moves forward. I also used cartoon visuals because I thought they would make the slides more interesting and easier for an 11-year-old to follow. I also learned that examples like an “ear weight” or “whisker weight” are just simple ways of explaining the idea. A real neural network learns from many different patterns in the data.

Working on this project also helped me understand how my topic connects with what the other group members researched. The network uses weights, bias, and activation functions to make a prediction, and then it checks how wrong that prediction was. Backpropagation helps figure out what needs to change, and gradient descent helps update the weights and bias so the network can do better next time. My contribution was researching my topic, creating my two slides, and helping with the overall conclusion slide. The biggest thing I learned from this project is that deep learning concepts are not as confusing when you break them down into simple examples and explain them step by step.

